Array Of Products

AlgoExpert · Medium · Arrays

The problem

- Out[i] = product of every element except a[i], without division.
- Prefix products then suffix products.

Intuition

- Each output is the product of everything to the left times everything to the right, so two sweeps replace division.

Approach

- 1. Fill each slot with the running product from the left
- 2. Sweep from the right with a second running product
- 3. Multiply it in
- 4. No division needed

Java solution

```java
public class ArrayOfProducts {
    public static int[] arrayOfProducts(int[] a) {
        int[] out = new int[a.length];
        int run = 1;
        for (int i = 0; i < a.length; i++) { out[i] = run; run *= a[i]; }
        run = 1;
        for (int i = a.length - 1; i >= 0; i--) { out[i] *= run; run *= a[i]; }
        return out;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n) output

Recap

- Each output is the product of everything to the left times everything to the right, so two sweeps replace division.
- Complexity: O(n)